Gas Sensor Classification Pipeline Documentation

Datathon Methodology and Code Documentation

Overview

This pipeline classifies gas types from sensor measurements using feature engineering, PCA dimensionality reduction, batch drift correction, and a weighted ensemble of Logistic Regression models. The objective is to remain robust to sensor drift and batch-specific distribution changes.

Feature Engineering

Raw sensor readings are transformed into statistical descriptors including mean, standard deviation, minimum, maximum, range, median, skewness, kurtosis, quartiles, and interquartile range. Multiple concentration transformations are generated to help the model capture non-linear concentration effects.

PCA

Principal Component Analysis compresses highly correlated sensor measurements into 20 principal components. Experimental testing found 20 components to outperform larger and smaller values. These components are appended to the engineered feature space.

Batch Drift Correction

Two drift-corrected representations are generated. Batch Standardization removes batch-specific mean and variance. Batch Centering removes batch-specific offsets while preserving scale relationships.

Model Architecture

Three Logistic Regression pipelines are trained independently using raw engineered features, batch-standardized features, and batch-centered features. StandardScaler is used within each pipeline to ensure comparable feature scales.

Ensemble Strategy

The three models output class probabilities which are blended using weighted averaging: 30% Raw, 50% Batch Standardized, and 20% Batch Centered. This ensemble improves robustness by combining complementary representations of the data.

Experimental Evaluation

Multiple alternatives were tested, including CatBoost, XGBoost, different PCA component counts, and additional drift correction methods. The Logistic Regression ensemble provided the strongest observed performance.

Output

The pipeline predicts a gas class for each measurement and exports results in the competition submission format.